SHRUTI SHARMA

B.Tech Data Science Engineering, IIT Mandi

Uttar Pradesh, India | 9389038285 | shrutisharma260905@gmail.com | github.com/shrutisharma26 | Portfolio

TECHNICAL SKILLS

Languages: Python, C++, Java (familiar), JavaScript, HTML, CSS
Full-Stack Development: HTML5, CSS3, JavaScript, Responsive Design, REST APIs, Flask, Vercel
AI/ML: scikit-learn, OpenCV, DeepFace, TF-IDF, Machine Learning, Reinforcement Learning, NLP, LLMs
App & Visualization: Streamlit, Plotly, REST APIs
Tools & DevOps: Git, GitHub, GitHub Actions (CI/CD), SQL, Streamlit Cloud
Core CS: Data Structures & Algorithms, OOP, Software Design, Problem Solving
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Organisation

PROJECTS

NewsLens – Hyper-Personalization Engine *2nd Place, HACK60 (IIT Mandi × HCLTech)*
Python, Reinforcement Learning, TF-IDF, LLMs, OpenCV

- Engineered a hybrid recommender system combining Thompson Sampling reinforcement learning, TF-IDF, and collaborative filtering to personalize news feeds for users at scale.
- Trained and validated the recommendation pipeline on 156K interactions from the industry-standard MIND dataset, improving relevance of ranked content.
- Built a real-time LLM re-ranking module (Groq / Llama 3) with OpenCV-based gaze tracking to incorporate live mood detection into content ordering.
- Optimized the end-to-end inference pipeline to achieve sub-500ms latency, 4x faster than the target benchmark, enabling real-time personalization.

AI Mock Interview System
Python, Streamlit, NLP

- Built a resume-aware interview question generator that parses candidate resumes and dynamically generates role-relevant questions using NLP.
- Implemented TF-IDF and cosine-similarity based answer scoring to automatically evaluate response quality against expected answers.
- Designed a modular four-stage pipeline: resume parsing → question generation → answer evaluation → automated PDF report generation.
- Deployed the application on Streamlit Cloud, enabling public, real-time access for interview practice.

Resume ATS Analyzer
Python, Streamlit, scikit-learn

- Developed an NLP-based scoring engine in scikit-learn to rank resumes against job descriptions and automatically flag missing skills.
- Built an interactive Plotly dashboard to visualize match scores, with exportable match reports for candidates.
- Deployed the tool as a Streamlit web app, allowing recruiters and candidates to interactively compare resumes against job descriptions.

AI Customer Query Classifier & FAQ Bot
Python, scikit-learn, Flask, SQL

- Built an NLP text-classification pipeline (TF-IDF + Logistic Regression) to categorize customer support queries into billing, technical, and general intents, evaluated with a held-out test split.
- Designed a SQL (SQLite) schema to log classified tickets and store FAQ data, and wrote analytics queries for category distribution and low-confidence-ticket flagging.
- Built a Flask REST API that serves real-time classification results with matched FAQ suggestions.

Student Performance Prediction API
Python, scikit-learn, Flask, GitHub Actions

- Trained a Random Forest classifier achieving 87% test accuracy to predict pass/fail outcomes from study hours, attendance, and past performance.
- Built and containerized a Flask REST API (gunicorn) to serve real-time predictions, deployable to Render/Azure/AWS.
- Set up a CI pipeline with GitHub Actions to automatically retrain and test the model on every push.

EDUCATION

Indian Institute of Technology (IIT) Mandi Himachal Pradesh, India
B.Tech in Data Science Engineering 2024 – Present

Wisdom Public School, Aligarh (CBSE) Uttar Pradesh, India
Class XII – 94.2%
Class X – 90.5%

ACHIEVEMENTS

- 2nd Place, HACK60 – Hackathon organized by IIT Mandi in collaboration with HCLTech.
- 2nd Place, Inter IIT 8.0 – Dance.